

Amirhossein Javaheri

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🌐 Webpage in LinkedIn 🐙 Github 🎓 Scholar

Professional Summary

PhD candidate in Electronic and Computer Engineering at the Hong Kong University of Science and Technology, specializing in graph representation tools for signal processing, machine learning, and financial analytics. Extensive research background includes work on compressed sensing and feature engineering during MSc and BSc studies, with proven contributions to image processing and speech recognition. My research has been published in leading international journals and conferences. Beyond academia, I have several years of industry experience in programming and algorithm development for computer vision and deep learning.

Interests

- Machine Learning
- Signal Processing
- Financial Engineering
- Data Analysis
- Optimization Techniques
- Statistical Inference
- Portfolio Optimization
- Deep Learning
- Computer Vision
- Inverse Problems

Experience

NOVE , Machine Learning Engineer

Tehran
Aug 2019 - Mar 2022

- Enhanced reliability of automatic algorithms for GPS-denied navigation to over 85% using aerial video recordings for autonomous localization in GNSS-deprived environments
- Developed computer vision algorithms achieving 50% faster performance for real-time implementation on embedded devices (NVIDIA Jetson Nano, Raspberry Pi)
- Developed deep learning framework for acoustic sneeze detection achieving over 95% accuracy to enhance COVID-19 health monitoring applications
- Created vision-driven AI activity recognition system on Instagram utilizing screen recording analysis
- Utilized Python (OpenCV, Tensorflow, Keras, Pytorch), C/C++, QT, GitLab, CUDA

Multimedia and Signal Processing Lab , Graduate Researcher

Tehran
Sep 2016 - May 2019

- Developed novel low-rank and sparse representation algorithms with faster run-time and higher accuracy for image inpainting
- Designed convex perceptual image quality assessment criteria that can help increase the robustness of image completion with lower computational complexity compared to the benchmark
- Utilized MATLAB, Python (Numpy, SciPy, Scikit-learn)

TCI Telecommunication Company, Summer Intern

Tehran
Jul 2014 - Oct 2014

- Developed and simulated a computer model in OPNET for monitoring communication networks with wireless routing-scheduling agents
- Utilized OPNET, NS-3

Skills

Programming: MATLAB, R, Python (OpenCV, Tensorflow, Keras, Pytorch, Scikit-learn), C/C++, HTML, SQL

Development: Visual Studio (Visual Basic), Jupyter, Git, Simulink, Qt, Docker

Software: L^AT_EX, Microsoft Office (Excel, Word, Powerpoint), Photoshop, AutoCAD

Language

- **English:** Fluent, TOEFL (94/120), Duolingo (135/160)
- **Persian (Farsi):** Native
- **Arabic:** Beginner

Education

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|------------|--|---------------------|
| PhD | Electronic and Computer Engineering
Hong Kong University of Science and Technology | Sep 2022 - Jul 2026 |
| | <ul style="list-style-type: none">• Thesis: Robust Graphical Modeling for Machine Learning and Financial Engineering• Supervisor: Prof. Daniel P. Palomar• GPA: 4.1/4.3 | |
| PhD | Electrical Engineering, Communications
Sharif University of Technology | Sep 2019 - Dec 2024 |
| | <ul style="list-style-type: none">• Thesis: Learning Graph-Structured Models from Incomplete Data• Supervisor: Prof. Farokh Marvasti• GPA: 19.55/20 | |
| MSc | Electrical Engineering, Communications
Sharif University of Technology | Sep 2014 - Sep 2016 |
| | <ul style="list-style-type: none">• Thesis: Low-Rank and Sparse Representations Learning for Image Inpainting• Supervisor: Prof. Farokh Marvasti• GPA: 18.37/20 | |
| BSc | Electrical Engineering
Sharif University of Technology | Sep 2010 - Sep 2014 |
| | <ul style="list-style-type: none">• Thesis: Speech Emotion Recognition Using Teager Energy Operator Features• Supervisor: Prof. Mohammad Bagher Shamsollahi• GPA: 17.63/20 | |

Honors and Awards

- Member of the National Elites Foundation (BMN), since 2010.
- Member of the National Organization for Development of Exceptional Talents (NODET), 2003-2010.
- Ranked 1st in high school among students of all branches with highest GPA, NODET High School, 2009.
- Ranked 3rd in the National Physics Olympiad, 2009.
- Ranked among the top 0.1% of (more than 280,000) participants in the Nationwide Undergraduate Entrance exam (Konkoor), 2010.
- Awarded honorary admission to the undergraduate program at the Electrical Engineering Department of Sharif University of Technology, 2010.
- Ranked 13th (among more than 18000 students) in the Nationwide Postgraduate Entrance exam in Electrical Engineering, Communications Major, 2014.

- Awarded honorary admission to the graduate program at the Electrical Engineering Department, Sharif University of Technology, 2013.
- Recipient of the Outstanding BSc Thesis Award, Department of Electrical Engineering, Sharif University of Technology, Spring 2014.
- Ranked 4th out of nearly 30 MSc graduate students of class 2015 with major in Communications Systems, Sep. 2015
- Honorary admission to the PhD program at the EE Department, Sharif University of Technology, June 2018.
- Ranked 18th out of nearly 1000 entrants of the PhD entrance exam, Apr. 2019.
- Ranked 1st among nearly 40 PhD students of class 2019 with a GPA of 19.55/20, EE department of Sharif University of Technology, June 2021.
- Recipient of the dual PhD fellowship in Electronic and Computer Engineering, jointly offered by HKUST and Sharif University of Technology, June 2022.
- Recipient of the Student Travel Grants from HKUST and the European Association of Signal Processing (EURASIP), 2024-2025.

Projects

Mathematical Graph Plotter with Syntax Parser

- Developed a program for plotting mathematical graphs, *Part of the C++ Programming Course Project, Under the Supervision of Dr. Shamsaie.*
- Tools Used: Visual Studio, C++

Target Following Robot with LEGO™ Mindstorms

- Designed a closed-loop control system to implement a robot that can follow light source, *Part of the Linear Control Course Project, Under the Supervision of Dr. Amin Nobakhti.*
- Tools Used: EV3 App

Automatic Speech Emotion Recognition

- Designed a GUI for detecting emotion (anger, happiness, anxiety, etc.) from recorded speech using artificial neural networks based on augmented features extracted via the Teager energy operator, *Part of the BSc Thesis Project, Under the Supervision of Dr. Mohammad Bagher Shamsollahi*
- Tools Used: MATLAB, Simulink

Adaptive Channel Estimation and Equalization

- Designed a an algorithm based on adaptive filters for channel estimation and equalization for error correction and distortion/noise reduction in wireless communication, *Part of the Adaptive Filters Course Project, Under the Supervision of Dr. Massoud Babaie-zadeh*
- Tools Used: MATLAB

Facial Recognition and Classification

- Developed a python program for classification of facial images using linear discriminant analysis (LDA), *Part of the Statistical Learning Course Project, Under the Supervision of Dr. Hoda Mohammadzade*
- Tools Used: Python

Automatic Hand Gesture Recognition

- Designed a game based on automatic hand gesture recognition (rock, paper, scissors) *Part of the Multi-Camera Vision course under the supervision of Dr. Hamid Aghajan*
- Tools Used: Python, OpenCV

Sparse Representation and Matrix Completion for Image Restoration

- Proposed and implemented fast and visually efficient methods based on novel image quality assessment criteria for completing images with missing regions (inpainting) *Part of the MSc Thesis Project, Under the Supervision of Dr. Farokh Marvasti*
- Tools Used: MATLAB, C (mex)
- github.com/javaheriamirhossein/CSIM-ALM 

Automatic Image Captioning

- Developed a Jupyter notebook for generating captions (descriptions) for images based on deep neural networks (LSTM and ResNet-50), *Part of Deep Learning Course Project, Under the Supervision of Dr. Emad Fatemizadeh*
- Tools Used: Jupyter Notebook, Python (Tensorflow, Keras)

AI Poem Generation

- Developed a python program for generating artificial poems using RNNs and word embedding models (GloVe), *Part of Deep Learning Course Project, Under the Supervision of Dr. Emad Fatemizadeh*
- Tools Used: Python (Pytorch)

Automatic Street-View House Number Detection

- Developed a Jupyter notebook for automatic detection of house numbers from street view (e.g., Google street view) images using CNNs, *Part of Computer Vision Course Project, Under the Supervision of Dr. Hoda Mohammadzadeh*
- Tools Used: Jupyter Notebook, Python (Tensorflow)
- github.com/javaheriamirhossein/House-number-detection 

Video Synopsis (Summerization) Using OpenCV Python

- Developed a python application for motion-based video synopsis. This application summarizes events (movements) in a video and outputs a short video displaying events in a chronological order. It also objects and their colors, calculates the speed of moving objects, counts the number of objects moving up and down, etc., *Part of the Computer Vision Course Project, Under the Supervision of Dr. Hoda Mohammadzadeh*
- Tools Used: Python (OpenCV)
- github.com/javaheriamirhossein/Video-Synopsis 

Graph Topology Learning From Incomplete Time-Series

- Proposed efficient methods (with 20-30% lower error) for (undirected/directed) graph learning and signal recovery from corrupted spatio-temporal data. These methods can be applied for prediction and imputation of multi-variate time-series (e.g., stock returns in a financial market) *Part of PhD. Research Project, Under the Supervision of Prof. Farokh Marvasti and Prof. Daniel P. Palomar*
- Tools Used: MATLAB, R
- github.com/javaheriamirhossein/STSRGL 

Graph-Based Representation and Clustering of Financial Data

- Proposed efficient algorithms for clustering of financial data with 10-20% higher accuracy. These clustering methods can be used in portfolio optimization and fraud detection, *Part of PhD. Research Project, Under the Supervision Prof. Daniel P. Palomar*
- Tools Used: R
- github.com/javaheriamirhossein/balanced-fingraph 
- github.com/javaheriamirhossein/bipartite-fingraph 

Learning Time-Varying Graph Models for Financial Data

- Novel algorithms proposed for time-varying (dynamic) graph learning from financial data that improve accuracy and speed by 10-30%). These clustering methods can be used in dynamic portfolio optimization achieving 5-10% higher Sharpe ratios *Part of PhD. Research Project, Under the Supervision of Prof. Daniel P. Palomar*
- Tools Used: R, MATLAB
- github.com/javaheriamirhossein/k-tvgraph 

Kalman Filtering for Financial Time-Series Analysis

- Implemented Kalman filtering method in financial applications including pairs trading and capital asset price modeling *Part of the Course Project Under the Supervision of Prof. Ling Shi*
- Tools Used: Python, Jupyter Notebook
- github.com/javaheriamirhossein/Kalman_in_Finance 

Selected Courses

Undergraduate/Graduate Courses

- **Introduction to C++ Programming** *Dr. Shamsaie, 20/20 (A+)*
- **General Mathematics 1** *Dr. Mohammad Ali Najafi, 17.2/20 (A)*
- **General Mathematics 2** *Dr. Reza Moghaddasi, 17.5/20 (A)*
- **Differential Equations** *Dr. Ali Salehi, 19.3/20 (A+)*
- **Numerical Computation** *Dr. Negin Bagherpoor, 19.3/20 (A+)*
- **Engineering Mathematics** *Dr. Mohammad Mobed, 18.6/20 (A)*
- **Electrical Circuit Theory** *Dr. Mahtab Mirmohseni, 20/20 (A+)*
- **Signals and Systems** *Dr. Massoud Babaie-zadeh, 17.5/20 (A)*
- **Linear Algebra** *Dr. Saleh Tavazoei, 18.9/20 (A+)*
- **Data Networks** *Dr. Mohammad Reza Pakravan, 17.5/20 (A)*
- **Adaptive Filters** *Dr. Massoud Babaie-zadeh, 17.8/20 (A)*
- **Information and Coding Theory** *Dr. Mohammad Reza Aref, 17.3/20 (A)*
- **Advanced Communication Systems** *Dr. Masoumeh Nasiri Kenari, 18.7/20 (A+)*
- **Digital Signal Processing II** *Dr. Farokh Marvasti, 19.1/20 (A+)*
- **Data Compression** *Dr. Farokh Marvasti, 19.4/20 (A+)*
- **Statistical Learning** *Dr. Hoda Mohammadzade, 20/20 (A+)*
- **Theory of Machine Learning** *Dr. Mohammad-ali Maddah-ali, 20/20 (A+)*
- **Deep Learning** *Dr. Emad Fatemizadeh, 18.8/20 (A+)*
- **Graphical Models** *Dr. Arash Amini and Dr. Mehdi Molkaraie, 19.4/20 (A+)*
- **Computer Vision** *Dr. Hoda Mohammadzadeh, 20/20 (A+)*
- **Convex Optimization** *Prof. Daniel P. Palomar (HKUST), A+*
- **Introduction to Networked Sensing, Control and Estimation** *Prof. Ling Shi (HKUST), A+*

Online Courses

- **Deep Learning Explained** at [edX.org](https://www.edx.org) 
- **C++ Tutorial** at [SoloLearn.com](https://www.sololearn.com) 
- **Python 3 Tutorial** at [SoloLearn.com](https://www.sololearn.com) 

Publications

Refereed

- A. Javaheri, H. Zayyani and F. Marvasti, "**Recovery of Missing Samples Using Sparse Approximation via A Convex Similarity Measure**", *SAMPTA 2017, Tallinn, Estonia, Jul 2017* [🔗](#).
- A. Javaheri, H. Zayyani and F. Marvasti, "**Sparse Recovery of Missing Image Samples Using a Convex Similarity Index**", *Elsevier Signal Processing* [🔗](#), vol. 152, May 2018.
- A. Javaheri, H. Zayyani, F. Marvasti and M.A.T. Figueiredo, "**Robust Sparse Recovery in Impulsive Noise via Continuous Mixed Norm**", *IEEE Signal Processing Letters* [🔗](#), vol. 25, Jun 2018.
- H. Zayyani, A. Javaheri, "**A Robust Generalized Proportionate Diffusion LMS Algorithm for Distributed Estimation**", *IEEE Transactions on Circuits and Systems II: Express Briefs* [🔗](#), doi: 10.1109/TCSII.2020.3029780, Oct 2020.
- M Delbari, A Javaheri, H Zayyani, F Marvasti, "**Non-coherent DOA Estimation via Majorization-Minimization Using Sign Information**", *IEEE Signal Processing Letters* [🔗](#), doi: 10.1109/LSP.2022.3162153, Mar 2022.
- A. Javaheri, A. Amini, F. Marvasti, and Daniel P. Palomar, "**Learning Spatio-Temporal Graphical Models From Incomplete Observations**", *IEEE Transactions on Signal Processing* [🔗](#), doi: 10.1109/TSP.2024.3354572, Mar 2024.
- A. Javaheri, J. V. De M. Cardoso, and D. P. Palomar, "**Graph Learning for Balanced Clustering of Heavy-Tailed Data**", *2023 IEEE 9th International Workshop on Computational Advances in Multi-Sensor Adaptive Processing (CAMSAP), Herradura, Costa Rica, pp. 481-485, Dec 2023* [🔗](#).
- A. Javaheri, A. Amini, F. Marvasti, and Daniel P. Palomar, "**Joint Signal Recovery and Graph Learning from Incomplete Time-Series**", *ICASSP 2024, Seoul, South Korea, Apr 2024* [🔗](#).
- A. Javaheri and D. P. Palomar, "**Learning Time-Varying Graphs for Heavy-Tailed Data Clustering**", *2024 European Signal Processing Conference (EUSIPCO 2024), Lyon, France, Aug 2024* [🔗](#).
- A. Javaheri, J. Ying, D. P. Palomar, and F. Marvasti, "**Time-Varying Graph Learning for Data with Heavy-Tailed Distribution**", *IEEE Transactions on Signal Processing* [🔗](#), doi: 10.1109/TSP.2025.3588173, Jul 2025.
- A. Javaheri and D. P. Palomar "**Clustering of Incomplete Data via a Bipartite Graph Structure**", <https://arxiv.org/abs/2505.08594> [🔗](#), March 2025, to appear in EUSIPCO 2025 Conference proceedings.

Preprints

- A. Javaheri and M. B. Shamsollahi, "**On Higher Order Positive Differential Energy Operator**", *arXiv preprint arXiv:1701.03834* [🔗](#), Jan 2017.
- A. Hippert-Ferrer, A. Sportisse, A. Javaheri, M. N. El Korso, and D. P. Palomar, "**Missing Data in Signal Processing and Machine Learning: Methods, Models, and Modern Approaches**", <https://arxiv.org/abs/2506.01696> [🔗](#), May 2025 (under review at IEEE Signal Processing Magazine).

Working Papers

- A. Javaheri and D. P. Palomar, "**A Bipartite Graph Model for Robust Clustering of Financial Data**", (to be submitted to IEEE Transactions on Signal and Information Processing Over Networks).
- A. Javaheri and D. P. Palomar, "**A Graph-Structured VAR Model for Data with Higher Order Temporal Dependencies and Heavy Tails**", (to be submitted to IEEE Signal Processing Letters).

Teaching

- **Teaching Assistant**, Optimization in FinTech, *Prof. Daniel P. Palomar, Spring 2025.*
- **Teaching Assistant**, Convex Optimization, *Prof. Daniel P. Palomar, Fall 2024.*
- **Teaching Assistant**, Data Driven Portfolio Optimization, *Prof. Daniel P. Palomar, Spring 2023, Spring 2024.*
- **Teaching Assistant**, Graph Signal Processing, *Dr. Arash Amini. Spring 2021.*
- **Teaching Assistant**, Computer Vision, *Dr. Hoda Mohammadzade, Spring 2021.*
- **Teaching Assistant**, Graphical Models, *Dr. Mehdi Molkararie, Fall 2020.*
- **Teaching Assistant**, Data Compression, *Dr. Farokh Marvasti, Fall 2019, Spring 2020.*

- **Teaching Assistant**, DSP II, *Dr. Farokh Marvasti*, *Spring 2019, Spring and Fall 2020*.
- **Teaching Assistant**, Circuit Theory, *Dr. Mahtab Mirmohseni*, *Fall 2014*.
- **Teaching Assistant**, Engineering Mathematics, *Dr. Ali Banai*, *Spring 2015*.
- **Teaching Assistant**, Digital Communications, *Dr. Jawad Salehi*, *Spring 2016*.
- Teaching as **Physics Olympiad Teacher** in NODET schools, *2013 - 2018*.

Professional Service

- **Peer Review:**
 - IEEE Transactions on Signal Processing
 - IEEE Transactions on Signal and Information Processing Over Networks
 - IEEE Signal Processing Letters
 - Elsevier Signal Processing
 - Elsevier Digital Signal Processing
- **Talks:**
 - [SAMPTA Conference](#), Tallinn, Estonia, Jul 2017
 - Digital Signal Processing Seminar, Sharif University of Technology, Tehran, Nov 2019
 - [CAMSAP Conference](#), Herradura, Costa Rica, Dec 2023
 - [ICASSP Conference](#), Seoul, South Korea, Apr 2024
 - [EUSIPCO Conference](#), Lyon, France, Aug 2024
 - Departmental Seminar, HKUST, Hong Kong, Dec 2024
 - [Future Leaders PG Seminar](#) in Data Science and AI, HKUST, Hong Kong, Apr 2025
 - [EUSIPCO Conference](#), Palermo, Italy, Sep 2025